PRACTICAL COURSE WORKBOOK

Climate Data, Tech & Sustainability

Use evidence to explain a public-interest problem

LEVEL	Beginner
GUIDED TIME	5 hours across 12 lessons
PROJECT	an evidence brief with a reproducible analysis
SKILLS	Climate data, Python, ESG, Carbon, Sustainability tech, Critical evaluation
REVIEWED	2026-08-25

WHAT YOU WILL BE ABLE TO DO

- Use Climate data appropriately in a realistic, bounded task.
- Use Python appropriately in a realistic, bounded task.
- Use ESG appropriately in a realistic, bounded task.
- Use Carbon appropriately in a realistic, bounded task.

WHO THIS IS FOR

Public-interest learners using evidence and technology responsibly.

BEFORE YOU BEGIN

- Basic research and spreadsheet skills

Use this guide beside the interactive lessons. Keep inputs, decisions, tests, limitations and the next improvement.

COURSE MAP

From foundation to finished work

Climate Data, Tech & Sustainability is a structured, practical course covering Climate data, Python, ESG, Carbon, Sustainability tech. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 Public problems	1. Stakeholders with Climate data 2. Institutions with Python 3. Evidence with ESG
02 Digital methods	1. Open data with Carbon 2. Services with Sustainability tech 3. Participation with Climate data
03 Policy design	1. Options with Python 2. Impact with ESG 3. Implementation with Carbon
04 Accountability	1. Measurement with Sustainability tech 2. Transparency with Climate data 3. Maintenance with Python

HOW TO STUDY EACH LESSON

1. Read the objective and identify the decision it supports.
2. Reproduce the worked example with the stated input.
3. Test one normal case and one boundary or failure case.
4. Complete the knowledge check without guessing.
5. Save a short evidence note: result, limitation and next action.

CAPSTONE PROJECT

Produce a reviewable Climate Data, Tech & Sustainability project using Climate data, Python, ESG.

DELIVERABLE	A working Climate Data, Tech & Sustainability artefact plus an evidence-based self-review.
TOOLS	A suitable Climate data environment, A plain-text decision log, Test data or realistic sample material

Production plan

1. Define the intended user, outcome and constraints.
2. Create the smallest complete result using Climate data.
3. Test one normal case, one boundary case and one failure response.
4. Revise the work from the evidence and preserve before-and-after results.
5. Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

Evidence log

INPUT	Original brief, data, requirements or test material.
DECISIONS	Chosen method and one reasonable alternative.
TESTS	Normal, boundary and failure results.
LIMITATION	What the result does not establish.
REVISION	One evidence-led improvement.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

Authoritative references

Open Government Data Toolkit

World Bank - reviewed 2026-08-21
https://opendatatoolkit.worldbank.org/en/data/opendatatoolkit/home

Digital government

OECD - reviewed 2026-08-21
https://www.oecd.org/en/topics/digital-government.html

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the Climate data scope.